

Clinical Pattern Recognition & HPI Cheat Sheet

This guide provides a standardized framework for ophthalmic technicians (COA, COT, COMT) to elevate clinical documentation beyond data entry into the realm of diagnostic support.

1. HPI Mastery: Bad vs. Elite Documentation

Transitioning from "standard" to "elite" requires documenting not just what the patient has, but what they *deny* in relation to their chief complaint. Use the following table to refine your HPI entries.

HPI Category	"Bad" (Basic) Documentation	"Elite" (Expert Scribe) Documentation
Ocular History	"Vision is blurry in the right eye."	"Pt reports sudden onset blurred vision OD; denies flashes, floaters, or 'curtain' over vision."
Medical History	"Patient has high blood pressure."	"Pt with history of HTN; denies recent fevers, chills, or neck stiffness associated with current headache."
Medication History	"Uses Latanoprost at night."	"Pt reports 100% compliance with Latanoprost QHS OU; denies drug allergies or recent use of systemic steroids."
Social History	"Patient lives alone."	"For current headache, pt denies recent head trauma or potential carbon monoxide exposure at home."
Chief Complaint	"Headache for two days."	"Pt reports 'worst headache of life' (thunderclap onset); denies dysarthria (slurred speech), paresthesia, or localized weakness."

Clinical Pearl: When a patient presents with a "thunderclap" headache or the "worst headache of their life," bypass the routine workup and immediately alert the ophthalmologist. This is a primary neurological red flag.

Essential Ophthalmic History Checklist

- **Actionable Abbreviations:** Use only approved clinical abbreviations (e.g., OD, OS, OU, NLP).
- **Systemic Manifestations:** Specifically document history of Diabetes Mellitus, Thyroid disease, and Autoimmune or Infectious diseases (HIV/AIDS, Tuberculosis).
- **Anatomy of History:** Ensure notes address relevant body systems (Respiratory, Cardiovascular, Endocrine, and Nervous).
- **Communication Log:** Document all interactions with the patient, family, and vendors to ensure a complete legal record.

2. Pattern Recognition: Retinal Detachment (RD) vs. Migraine

Differentiating between a mechanical retinal event and a systemic neurological event requires precise diagnostic tools and "Red Flag" history.

- Diagnostic Tools (RD Identification)**
- **Visual Assessment:** Record acuity for all levels, specifically including Counting Fingers (CF), Hand Motion (HM), Light Perception (LP), and No Light Perception (NLP).
 - **Ophthalmic Imaging:** Perform and record a **Scanning laser test for glaucoma/retina (HRT, OCT)**.
 - **Internal Photography:** Utilize Fundus photography to document retinal integrity.
 - **Pinhole Testing:** Use a pinhole occluder to determine if the vision change is refractive or structural.
- Red Flag History (Migraine/Headache)**

- **Neurological Indicators:** Specifically document the presence or absence of **dysarthria (slurred speech)** , paresthesia, or localized weakness.
- **Systemic Red Flags:** Ask about fevers, chills, or nuchal rigidity (neck stiffness).
- **Environmental Factors:** Screen for potential carbon monoxide exposure.
- **Severity:** Note if the headache was a "thunderclap" onset.

3. Pattern Recognition: Dry Eye vs. Conjunctivitis

Use assessment tasks and microbiology data to differentiate between surface dryness and infectious or allergic inflammation. | Condition | Key Assessment Task (Source 2) | Relevant Medication Category (Source 2) || ----- | ----- | ----- || **Dry Eye** | Tear tests: Schirmer, BUT (Break-up Time), and Rose Bengal. | Ocular lubricants. || **Bacterial Conjunctivitis** | Assist in collection of conjunctival/corneal specimens. | Anti-infective agents: **Anti-bacterial** . || **Viral Conjunctivitis** | Identify virus types/pathways of transmission. | Anti-infective agents: **Anti-viral** . || **Fungal Conjunctivitis** | Identify fungus/protozoan microorganisms. | Anti-infective agents: **Anti-fungal/Anti-parasitic** . || **Allergic Conjunctivitis** | Assess environmental triggers/inflammation. | **Allergy medications** . |

4. Pattern Recognition: Neurological vs. Ocular Diplopia

Diplopia must be categorized as either a mechanical ocular motility issue or a potential neurological emergency. **Neurological Red Flags (Immediate Referral)**

1. **Facial Droop & Dysarthria:** Classic indicators of neurological deficit.
2. **Gait Instability:** Unexplained difficulty walking or loss of balance.
3. **Confusion:** Any acute change in mentation.
4. **Localized Weakness:** Weakness confined to one side of the body or a specific limb.
5. **Syncope/Seizure:** History of fainting or tremors associated with vision changes. **Ocular Assessment Skills**

- **Versions and Ductions:** Evaluate the range of motion to identify muscle-specific anomalies.
- **Cover Tests:** Detect and identify the direction of a phoria (latent) vs. tropia (manifest).
- **Prism Testing:** Utilize Worth 4-Dot, Maddox Rod, or Krimsky tests.
- **Cranial Nerve Evaluation:** Assess function for **Cranial Nerves III, IV, V, VI, and VII** .
- **Convergence:** Measure the Near Point of Accommodation and Convergence.

5. Quick Reference: Vital Signs & Supplemental Diagnostics

Mandatory Vital Signs (Source 2)

- **Blood Pressure:** Essential for cardiovascular baseline.
- **Pulse:** Heart rate and rhythm.
- **Respiration:** Rate and quality of breathing. **Advanced Diagnostic Functions**
- **Scanning Laser (OCT/HRT):** For retina and glaucoma analysis.
- **Topography:** Mapping corneal surface curvature.
- **Angiography:** Identify phases in **Fluorescein Angiography (FA)** and **Indocyanine Green Angiography (ICG)** . **Clinical Pearl: The Lens Clock "Cycle-Up" Method** To determine the refractive index of an unknown material:

1. Read the power from a manual lensometry first (e.g., -5.00D).
2. Set the digital lens clock to the **1.49 index** and measure the front/back curves.
3. If the combined power does not match the lensometer, **cycle up** through the indices (1.53, 1.56, 1.59, 1.61, etc.) until the lens clock calculation matches the manual lensometry reading.

6. Technical Skills Checklist for the Ophthalmic Team

COA (Basic Level)

- **Standard Precautions:** Follow Universal Precautions and infection control to maintain clinical asepsis.
- **Automated Lensometry:** Neutralize spectacles using automated equipment.
- **Vital Signs:** Measure and record BP, pulse, and respiration.
- **Documentation:** Enter clinical data and physician dictation into EMR accurately.**COT**

(Intermediate Level)

- **Manual Lensometry:** Neutralize distance correction and bifocal/trifocal adds.
- **Retinoscopy:** Perform manual retinoscopy to find initial objective refractive error.
- **Visual Fields:** Perform automated perimetry (e.g., Humphrey Visual Field).
- **Keratometry:** Perform manual keratometry to measure corneal curvature.**COMT**

(Advanced Level)

- **Prism Mastery:** Identify and measure ground-in prism or slab-off using manual lensometry.
- **Advanced Imaging:** Perform Fundus Photography and identify specific phases of Fluorescein Angiography.
- **Complex Motility:** Measure deviations using prism and cover tests at distance; evaluate versions and ductions for abnormalities.
- **Lens Clock:** Utilize the Geneva lens clock to determine refractive index and base curves.